Week 5 Progress Writeup

1. **Each team member's contribution:**

Shreya Shetty - This week, I built Feature Agent 2B on top of Pranav's shared feature table, which filters it down to one verified, leakage-checked table per prediction model (Model 1: 7,208 mailed rows; Model 2: 254 redeemed TypeB/C rows; Model 3: 43,396 rows, 3,229 treated and 40,167 clean control households), with a safety rail that stops the pipeline if any model's answer column ever leaks into another model's inputs. On top of that, I built a baseline T-learner for Model 3 (uplift), trained and evaluated out-of-time on the real data — it confirmed campaign targeting is non-random (treated households spend ~4.6–4.75x more pre-campaign than control, consistent across multiple checks) and showed a directionally correct but not yet statistically significant uplift signal, an honest first-baseline result that points directly at next steps rather than overclaiming. I also restructured the repo (all documentation consolidated under docs/, agent code organized consistently under src/) and set up enforced PR-based branch protection now that multiple people are committing to the same codebase. At Friday's mentor meeting, we presented this work live, running the actual pipeline and baseline in real time rather than slides, and received direction to refine each model including my Model 3's outcome definition by distinguishing treated households who purchased because they redeemed the coupon from those who purchased the eligible product incidentally, without ever using it (this was also part of my main proposal for model 3), the latter group behaves identically to organic/control demand and shouldn't be counted as a campaign success. That refinement is this week's next concrete step. Separately, Anna picked up and merged Model 1 and Model 2 baselines using the same pattern as Model 3's, confirming the shared pipeline design works as a system across contributors, not just in isolation.

John - This week I focused on evaluation design for Model 1, the campaign response model. I reviewed Anna's Model 1 and 2 baselines in PR #8 and identified three checks to run before relying on the random forest's 61% top-20% capture rate. The first is comparing campaigns individually against a baseline that accounts for campaign type. The second is testing campaign by campaign over time, since the current test set contains only one Type A campaign. The third is confirming the predicted probabilities are calibrated for the Decision Lab's forecasts. I also outlined next steps to improve the model, including LightGBM, category-level offer-fit features, and per-household explanations, and planned the evaluation code so the Evaluation Agent can reuse it for Models 2 and 3. I presented this plan at Friday's mentor meeting, took meeting notes, and sent our mentors a follow-up email summarizing progress, feedback, and next week's plan. I also presented the worked example section of our Problem Description presentation in class.

Anna - This week, I built the Model 1 (response) and Model 2 (expected value) baselines on top of Feature Agent 2B's per-model tables, following the same pattern as the Model 3 baseline, so all three models now have a working baseline. I ran the full pipeline end to end on the real data, from data ingestion through the feature tables to both models, and evaluated each model out-of-time against the benchmarks in our evaluation plan, adding a simpler comparison model for each. On GitHub, I worked on a dedicated branch, updated the README with run commands and evaluation notes for the two new scripts, and opened PR #8 documenting the results and open assumptions for team review before merging it into main. The Model 1 baseline clearly outperforms the "redeemed before" rule (61% vs. 52% of redeemers reached in the top 20%), while Model 2 does not yet beat the "usual spend" benchmark given its small sample of 254 rows. I presented both baselines and their results at Friday's mentor meeting.

Pranav - This week I built step 2A of the Feature Agent, the shared feature table that all three models are built from. It takes Claudia's household × campaign table (75,000 rows) and adds, for each household and campaign, how the household shopped before the campaign started (weekly spend, trips, coupon use, past redemptions and spend on the campaign's products), along with each model's answer column. The result is one table of 75,000 rows and 47 columns that Shreya's step 2B then filters for each model. To make sure no column uses information from after a campaign starts, every column only looks at data from at least a week before the start, and the agent checks this on every run by scrambling all later data and confirming nothing changes. Before building it, I tested the candidate columns on the real data. Among other things, I found that how often a household uses coupons in general is the strongest predictor of redemption, even though it wasn't on our original list. I wrote up the design and the column definitions in the repo, added run instructions to the README and removed duplicate rows from the coupon table.

Zhiyan - This week I built and ran the Data Agent end to end on the full dunnhumby dataset. I started by loading all eight tables with a fixed format for each (column names, data types, null and duplicate counts) and saving them as Parquet files. While doing this I found that the demographics table's column names differ from the project PDF, so I wrote the real names into the loader.I then built the master table, with one row for each household–campaign pair. It has 75,000 rows (2,500 households × 30 campaigns), and the label `redeemed_flag` records whether the household used a coupon during that campaign. Of the 7,208 mailed pairs, 889 redeemed a coupon (12.3%). The rate differs by campaign type: 16.0% for Type A, 7.9% for Type B and 7.7% for Type C. I deliberately kept behavior summaries such as total spend out of this table, so the Feature Agent can build them using only data from before each campaign starts. That avoids data leakage. The table passed all eight of my self-checks.Next I wrote a script that checks the tables against each other and logs every finding. Across 34 checks, 23 passed, 7 were flagged and none failed, so the tables link up correctly. The flags were 5,164 duplicate coupon rows, 36 discounts with the wrong sign, 14,466 rows with quantity of zero or less, 18,850 rows with sales value of zero or less, 23,136 rows with quantity above 100, one campaign that ends after our data ends, and 15,245 conflicting display records.

2. Plan for upcoming week

Next week, we plan to refine each stage of the pipeline based on mentor feedback. For the Data Agent, we will make the data quality checks more data-driven, look more closely at the anomalies flagged so far, and start adding an LLM layer to help review flagged issues. For the Feature Agent, we will document and share the feature set and settle where feature selection fits in the pipeline. On the modeling side, we will evaluate the Model 1 baseline more rigorously and begin testing improvements, continue refining Model 2 given its small sample, and further explore whether current campaign targeting adds lift for Model 3. Throughout, we will keep the agents' outputs consistent so each model can move smoothly through the shared pipeline.